

Theorem (Infinitude of the primes). *For all $n \in \mathbb{N}$, there exists $p \in \mathbb{N}$ such that $p > n$ and p is prime.*

Proof. Take $n \in \mathbb{N}$, write $m := n! + 1$, and let p be a prime factor of m .

- Note that $m \neq 1$ because $n! \neq 0$.
- Therefore, p is prime.
- Suppose for contradiction that we do not have $p > n$.
- Then $p \leq n$.
- We have $p \mid m$ by definition of p .
- Note that $p > 0$ because p is a factor of m .
- Also, $p \mid n!$ because $p \leq n$ by assumption and $p > 0$.
- So $p \mid 1$ by modular arithmetic.
- Thus, $p = 1$.
- But 1 is not prime $\times$. □